

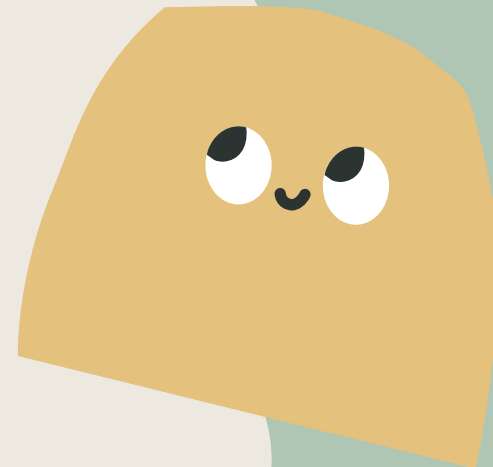
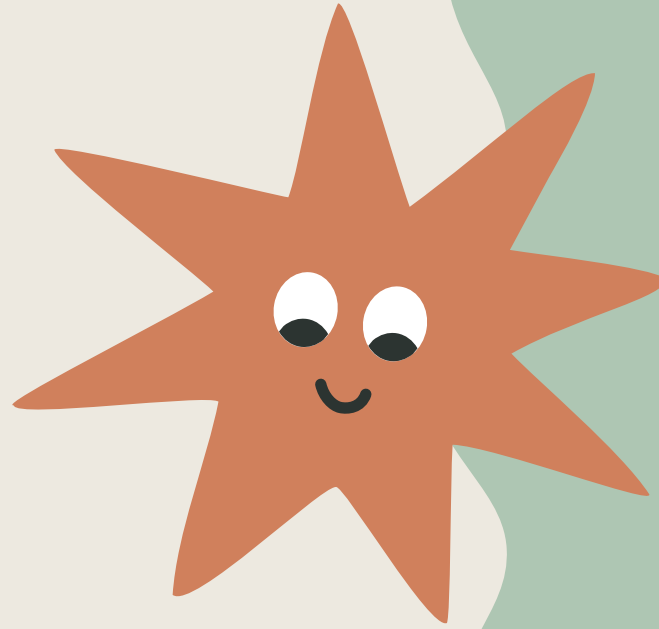
UNIT 1

FUNDAMENTALS OF

COMPUTER

9TH COMPUTER SCIENCE
SINDH TEXTBOOK BOARD

AWINASH GOSWAMI



Agenda

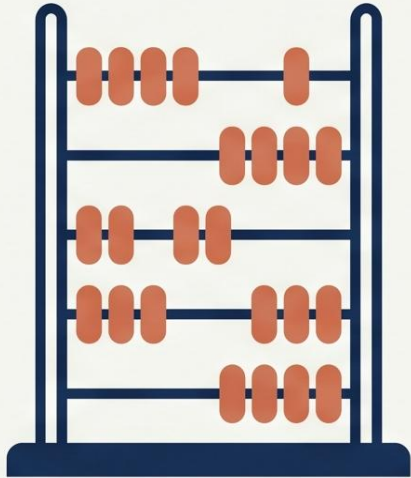
Evolution of Computers

Introduction to computer

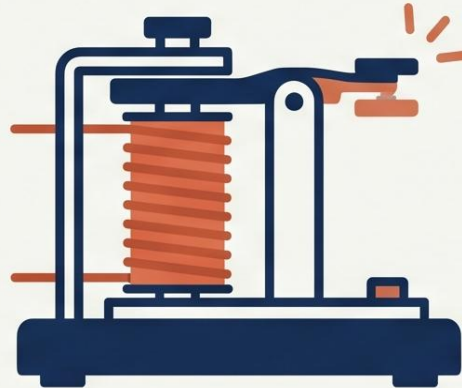
A computer is an electronic data processing machine.

EVOLUTION OF COMPUTERS

MECHANICAL ERA
(DARK AGE)



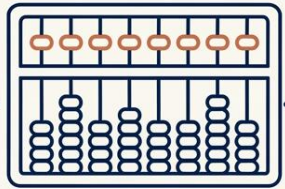
ELECTRO-MECHANICAL
ERA (MIDDLE AGE)



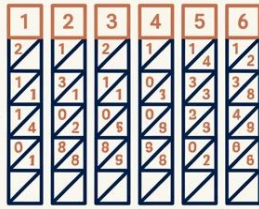
ELECTRONIC ERA
(MODERN AGE)



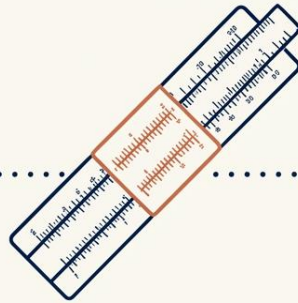
MECHANICAL ERA (DARK AGE)



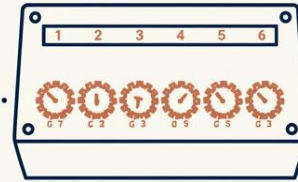
Abacus
(3000 B.C.)



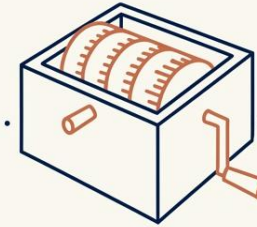
Napier's Bones
(1612 A.D.)



Slide Rule
(1622 A.D.)



Pascaline
(1642 A.D.)



Leibniz's Calculator
(1694 A.D.)



Babbage's Engine
(1822 & 1837 A.D.)

Abacus (3000 B.C.)

- Abacus was invented about 5000 years ago.
- It is also known as counting frame.
- Abacus is still used to teach basic arithmetic operations to the students.
- Abacus is considered as first computer prototype.



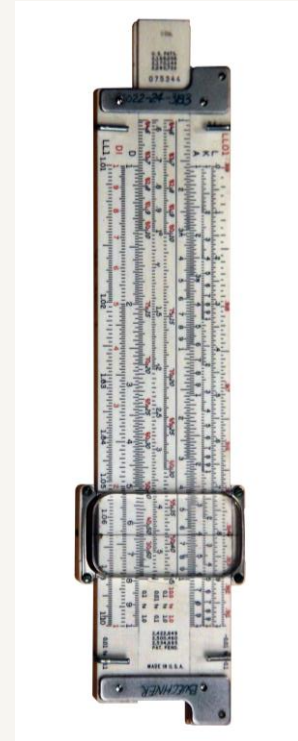
Napier's Bones (1612 A.D.)

- Scottish mathematician John Napier developed Napier's Bones, in 1612.
- It is also called Napier's Rods.
- It was a small machine that contained ten rods. The rods were made up of bones, wood or metal.
- Napier's Bones was used for multiplication and division.



Slide Rule (1622 A.D.)

- English mathematician William Oughtred invented a Slide Rule.
- This was basically a pair of two moveable rules placed side by side.
- They had a standardized set of markings called scales. These scales were used for mathematical computations.
- Slide Rule was used to multiply and divide numbers. The modern slide rules are still used in some countries



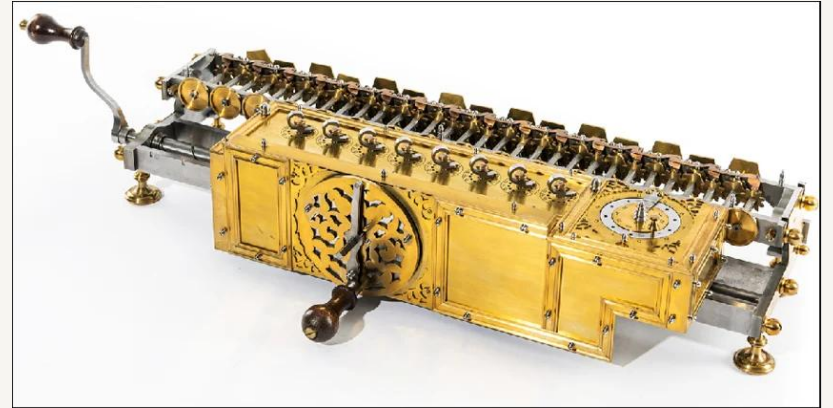
Pascaline or Pascal's Calculator (1642 A.D.)

- French mathematician Blaise Pascal invented a mechanical calculator which was called Pascaline.
- He developed this machine to help his father who was a tax collector.
- Pascaline can perform addition and subtraction by eight figures. User had to adjust wheels in order to perform these calculations.



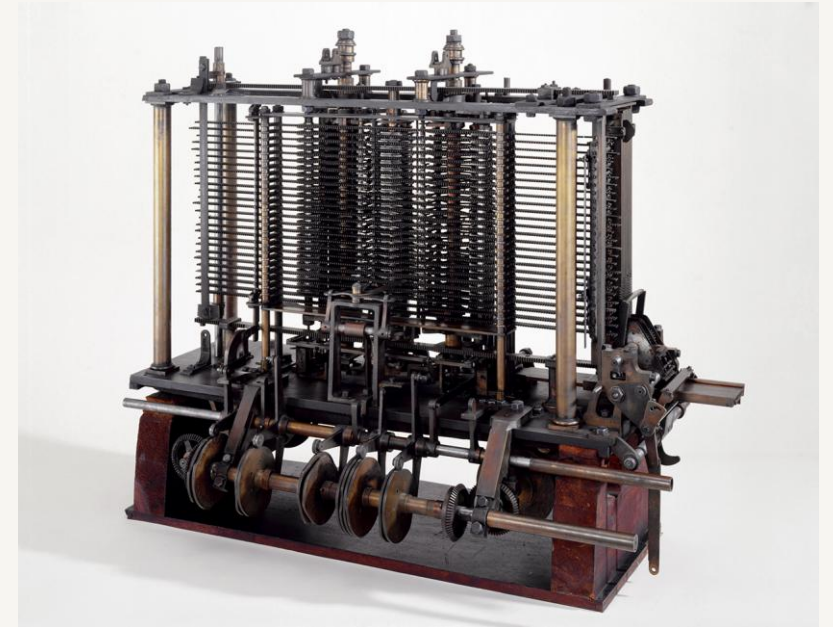
Leibniz's Calculator (1694 A.D.)

- Leibniz's Calculator was developed by the German mathematician Gottfried Wilhelm Leibniz.
- It was also called stepped reckoner.
- It was the first calculator that could perform all four basic arithmetic operations; addition, subtraction, multiplication and division



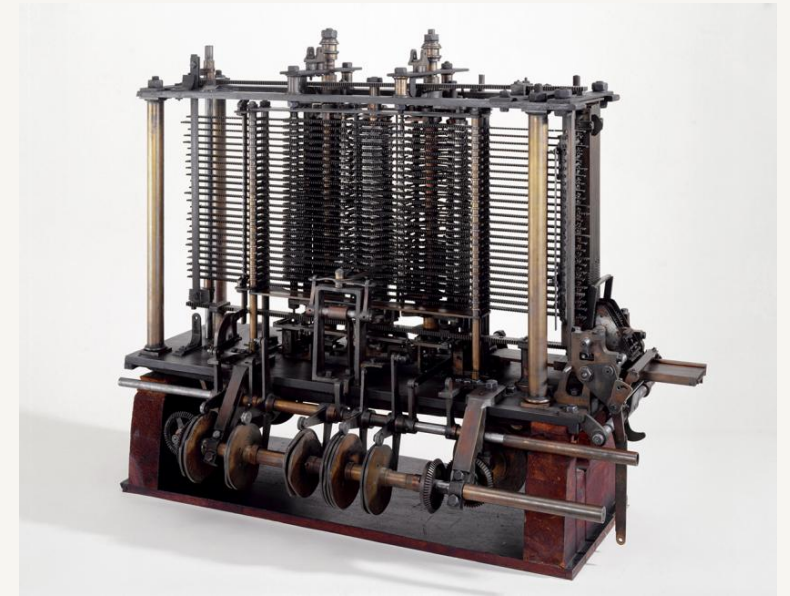
Babbage's Difference and Analytical Engines (1822 and 1837 A.D.)

- Charles Babbage, an English mathematician and engineer, is called the Father of Computers for creating the first complete computing machine.
- His first invention, the Difference Engine (powered by steam), was a large metal calculator that:
 - Stored data temporarily in mechanical memory.
 - Printed results from calculations using difference equations.

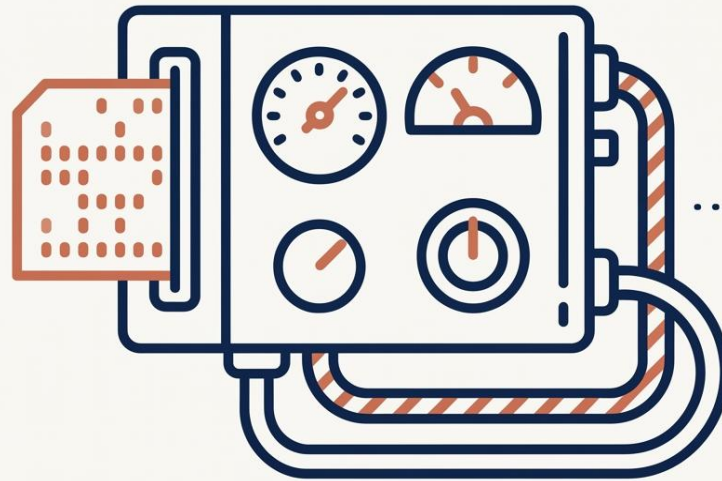


Babbage's Difference Cont....

- In 1837, he designed the Analytical Engine, which he never built. Its advanced design included:
 - An ALU (Arithmetic Logic Unit).
 - Programmatic flow control.
 - Punched-card programming and integrated memory.
- Historians view it as the first concept of a general-purpose computer due to its comprehensive design.



ELECTRO-MECHANICAL ERA (MIDDLE AGE)



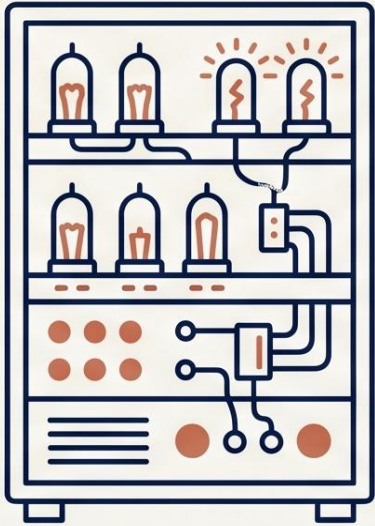
Hollerith's Tabulating Machine (1890 A.D.)

Herman Hollerith's Tabulating Machine (1890 A.D.)

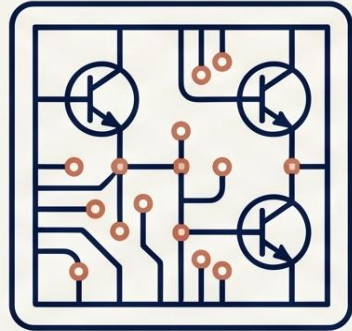
- In 1890, American inventor Herman Hollerith invented the Tabulating Machine. It was first used for the U.S. Census that year.
- While the 1880 census took 7 years to complete, Hollerith's machine finished the 1890 census in just 6 months—a huge speed boost.
- The machine became very popular and was widely used in many U.S. government offices.



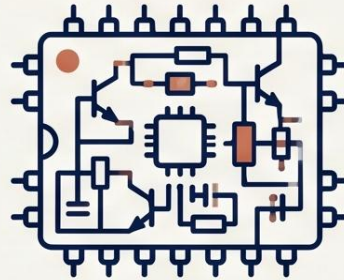
ELECTRONIC ERA (MODERN AGE)



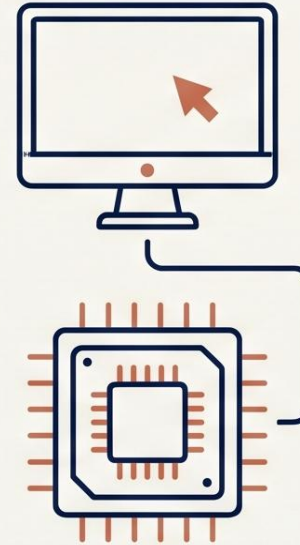
First Generation
(1940–1956)



Second Generation
(1956–1963)



Third Generation
(1964–1971)



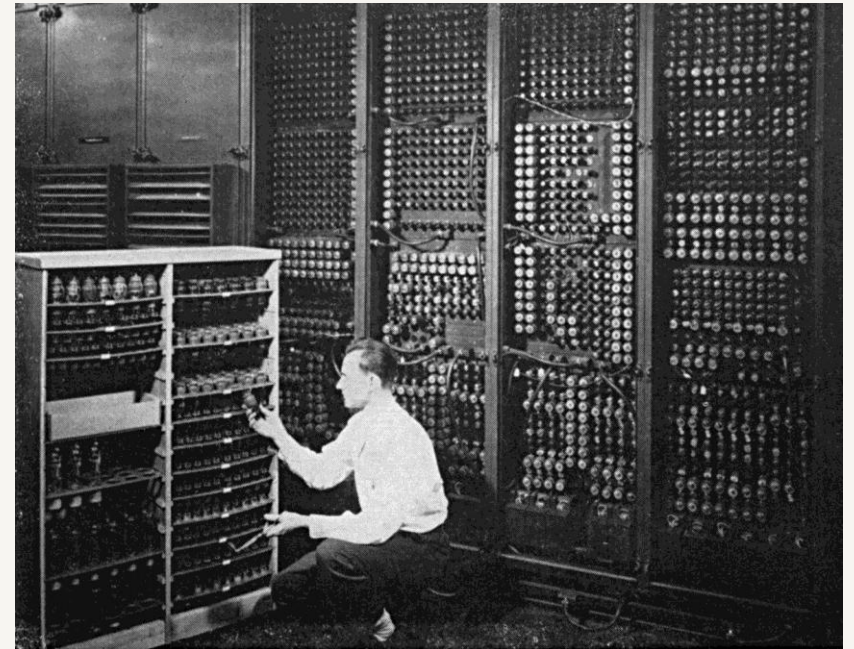
Fourth Generation
(1971–Present)



Fifth Generation
(Present and Beyond)

First Generation of Computers (1940 to 1956)

- First-generation computers (based on vacuum tubes) were very large in size. They used:
 - Machine language (1s and 0s) for programming.
 - Magnetic drums as primary internal storage.
 - Punched cards for input.
 - Batch processing operating systems.
- These computers were mainly used for scientific and research purposes.
Examples: ENIAC and UNIVAC.



Second Generation of Computers (1956 to 1963)

- Second-generation computers used transistors, making them smaller, faster, cheaper, and more efficient than first-generation machines.
- Key features:
 - Assembly language and high-level language FORTRAN were introduced.
 - Magnetic core replaced drums as primary internal storage.
 - Punched cards continued for input.
 - Batch processing and multiprogramming operating systems were used.
- These computers were used for commercial production, scientific, and engineering work.
Examples: IBM 7094 and IBM 1401.



Third Generation of Computers (1964 to 1971)

- Third-generation computers used Integrated Circuits (ICs), which further decreased size and increased speed and efficiency while making computers less expensive.
- Key features:
 - High-level languages like Pascal and COBOL were used.
 - Keyboard (input) and monitor (output) made computers easier to use.
 - Time-sharing and real-time operating systems were introduced.
 - Computers were now used for database management and automatic industrial control.
- Examples: IBM 360 and IBM 370.



Fourth Generation of Computers (1971 to Present)

- Fourth-generation computers used microprocessors, which was revolutionary. They became faster, cheaper, smaller, and more reliable.
- Key features:
 - Used semiconductor memories (RAM and ROM); magnetic storage became popular.
 - More high-level languages introduced – C, C++, Java, etc.
 - Used in almost every field – space, business, art, and more.
 - Operating systems: time-sharing, real-time, and distributed.
 - Saw the development of Graphical User Interfaces (GUIs) for easier use.
- Examples: Apple Macintosh, IBM PC.



Fifth Generation of Computers (Present and Beyond)

- Fifth-generation computers are still being developed. They aim to make computers capable of self-learning, reasoning, and generalization.
- Key features:
 - Will be able to process human languages and control machines like robots.
 - Based on Artificial Intelligence (AI) , which includes branches like Machine Learning, Deep Learning, Natural Language Processing, Robotics, Expert Systems
- These computers represent the future of intelligent and human-like computing.

